

Ryck et al, Neural Networks (2021), 143, 732-750 9/17

Sobolev space

P.734

$$W^{k,p}(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega), \forall \alpha \in \mathbb{N}_0^d, |\alpha| \leq k\}$$

$$L^p(\Omega) = \{f : \int_\Omega |f|^p d\tilde{x} < \infty\}$$

$$L^\infty(\Omega) = \{f : \sup_\Omega |f| < \infty\}$$

$$W^{k,\infty}(\Omega) = \{f \in L^\infty(\Omega) : D^\alpha f \in L^\infty(\Omega) \forall \alpha \in \mathbb{N}_0^d, |\alpha| \leq k\}$$

big O notation

$$f(\varepsilon) = O(\varepsilon), \text{ for small } \varepsilon. \Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{f(\varepsilon)}{\varepsilon} = C < \infty$$

Neural Networks (2021), 134, 107-130

Given activation function $\sigma(x)$.

Assume $\sigma^{(p)}(b) \neq 0$.

$$\frac{d}{dW} \sigma(WX+b) = X \sigma'(WX+b)$$

$$\frac{d^2}{dW^2} \sigma(WX+b) = X^2 \sigma''(WX+b)$$

$$\frac{d^p}{dW^p} \sigma(WX+b) = X^p \sigma^{(p)}(WX+b)$$

$$\Rightarrow \left. \frac{d^p}{dW^p} \sigma(WX+b) \right|_{W=0} = X^p \sigma^{(p)}(b)$$

$$\Rightarrow X^p = \frac{1}{\sigma^{(p)}(b)} \left(\left. \frac{d^p}{dW^p} \sigma(WX+b) \right|_{W=0} \right)$$

Finite difference method to approx. derivatives.

$$f(a+h) = f(a) + f'(a)h + \frac{f''(a)}{2}h^2 + \frac{f'''(a)}{6}h^3 + \frac{f^{(4)}(a)}{24}h^4 + O(h^5)$$

$$f(a-h) = f(a) - f'(a)h + \frac{f''(a)}{2}h^2 - \frac{f'''(a)}{6}h^3 + \frac{f^{(4)}(a)}{24}h^4 + O(h^5)$$

$$\Rightarrow f(a+h) - f(a-h) = 2f'(a)h + \frac{f'''(a)}{3}h^3 + O(h^5)$$

$$\Rightarrow \boxed{\frac{f(a+h) - f(a-h)}{2h}} = f'(a) + \frac{f'''(a)}{6}h^2 + O(h^4)$$

center difference to approx. $f'(a)$

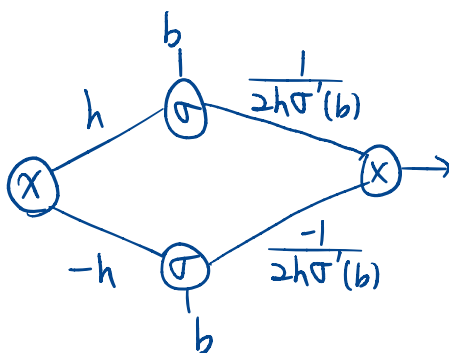
$$\frac{f(a+h) - 2f(a) + f(a-h)}{h^2} = f''(a) + O(h^2)$$

center difference to approx. $f''(a)$

ex: $p=1$, we want to approx. $\left. \frac{d}{dw} \sigma(wx+b) \right|_{w=0}$

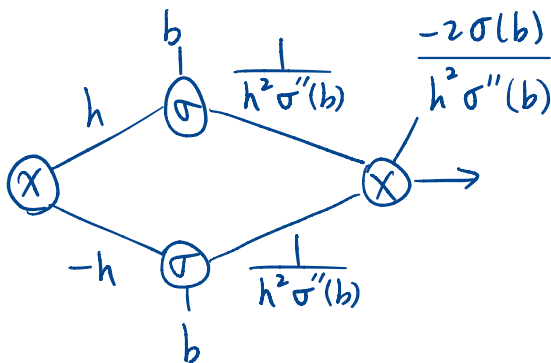
$$\left. \frac{d}{dw} \sigma(wx+b) \right|_{w=0} = \frac{\sigma((0+h)x+b) - \sigma((0-h)x+b)}{2h} + O(h^2)$$

$$\Rightarrow \chi = \frac{1}{\sigma'(b)} * \frac{\sigma(hx+b) - \sigma(-hx+b)}{2h} + O(h^2)$$



$$\left. \frac{d^2}{dw^2} \sigma(wx+b) \right|_{w=0} = \frac{\sigma(hx+b) - 2\sigma(b) + \sigma(-hx+b)}{h^2} + O(h^2)$$

$$\Rightarrow \chi^2 = \frac{1}{\sigma''(b)} * \frac{\sigma(hx+b) - 2\sigma(b) + \sigma(-hx+b)}{h^2} + O(h^2)$$

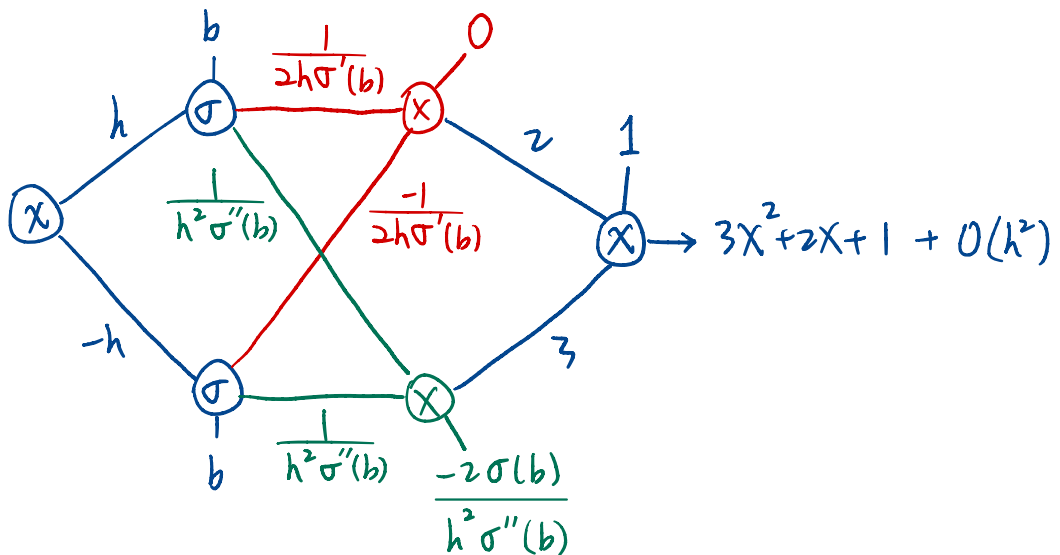


$$f_0(x) = (W_4 (W_3 \sigma(W_2 x + b_2) + b_3) + b_4) + O(h^2)$$

$$= \tilde{W} \sigma(W_2 x + b_2) + W_4 b_3 + b_4 + O(h^2)$$

Remark

To approximate $\{x^0, x^2, \dots, x^{2p}\}$, we need a neural network $1 \times 2p \times 2p$.



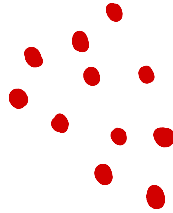
Classification

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group 0



group 1



data : $\{(x_i, y_i)\}$, $y_i \in \{0, 1\}$

hypothesis : $h_\theta(x)$

loss : MSE : $\frac{1}{N} \sum_i \|h_\theta(x_i) - y_i\|^2$

Cross-entropy : $y_i = h_\theta(x_i) + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$

Bernoulli distribution

let $Y \sim \text{Bernoulli}(p)$

$P(Y=1) = p$, $P(Y=0) = 1-p$, $0 \leq p \leq 1$.

Suppose we have a independent trial.

$\{y_1, \dots, y_n\}$. $y_i \in \{0, 1\}$.

probability mass function

$$f(y|p) = p^y (1-p)^{1-y}$$

$$f(y=1|p) = p^1 (1-p)^0 = p$$

$$f(y=0|p) = p^0 (1-p)^1 = 1-p$$

Likelihood function

$$L(p) = \prod_{i=1}^n p^{y_i} (1-p)^{1-y_i}$$

maximum likelihood estimation (MLE)

find p^* s.t. $L(p)$ is max.

$$p^* = \underset{p}{\operatorname{argmax}} L(p) = \underset{p}{\operatorname{argmax}} \ln(L(p))$$

$$\ell(p) = \ln(L(p)) = \sum_{i=1}^n y_i \ln p + (1-y_i) \ln(1-p)$$

$$\frac{d}{dp} \ln(L(p)) = \sum_{i=1}^n \frac{y_i}{p} - \frac{1-y_i}{1-p}$$

$$= \frac{1}{p} \sum y_i - \frac{n}{1-p} + \frac{1}{1-p} \sum y_i$$

$$= \frac{1}{p(1-p)} \sum y_i - \frac{n}{1-p}$$

$$\Rightarrow p^* = \frac{1}{n} \sum y_i$$

Assume that $p(y=1|x) = h_\theta(x)$

$$p(y=0|x) = 1 - h_\theta(x)$$

$$\Rightarrow f(y|x; \theta) = h_\theta^y(x) (1 - h_\theta(x))^{1-y}$$

$$L(\theta) = \prod_i h_\theta^{y_i}(x_i) (1 - h_\theta(x_i))^{1-y_i}$$

find θ s.t. $L(\theta)$ is max

$$\theta^* = \operatorname{argmax}_{\theta} (\ln L(\theta))$$

$$= \operatorname{argmin}_{\theta} (-\ln L(\theta))$$

$$= \operatorname{argmin}_{\theta} \sum_i \left(-y_i \ln h_\theta(x_i) - (1-y_i) \ln (1 - h_\theta(x_i)) \right)$$

cross entropy loss

Gradient descent method

$$\theta_{n+1} = \theta_n - \mu \nabla_{\theta} \text{Loss}$$

$$= \theta_n - \mu \left\{ \sum_i -y_i \frac{\nabla_{\theta} h_{\theta}(x)}{h_{\theta}(x)} - (1-y_i) \frac{-\nabla_{\theta} h_{\theta}(x)}{1-h_{\theta}(x)} \right\}$$

$$= \theta_n + \mu \left\{ \sum_i \frac{y_i - h_{\theta}(x)}{h_{\theta}(x) (1-h_{\theta}(x))} \nabla_{\theta} h_{\theta}(x) \right\}$$
